Unit 3 Look Back: Energy and Matter

Flex Page · Canvas Original #40 · Biology I and Advanced Biology I, Module 3

Name _____ Period _____ Date _____ Flex # _____

1. Rate yourself before you start

Target	I can...	Got it	Shaky	Lost
LT 3.1	explain how ATP releases energy and how the cell rebuilds it	○	○	○
LT 3.2	describe photosynthesis: inputs, outputs, where, and what changes its rate	○	○	○
LT 3.3	describe cellular respiration with and without oxygen, and who does it	○	○	○
LT 3.4	explain how the two processes connect: matter cycles, energy flows	○	○	○

2. Seven questions. Circle one answer for each.

1. (LT 3.4 · SOL Challenge Q27) A sealed glass jar holds a pond snail and a water plant, placed near a window. Weeks later both are still alive. Which exchange between them explains that?

(A) Both the plant and the snail release oxygen, so the jar never runs out
(B) The plant releases carbon dioxide and the snail releases oxygen
(C) Neither organism needs any gas, so the seal does not matter
(D) The plant releases oxygen the snail uses, and the snail releases carbon dioxide the plant uses

2. (LT 3.3 · SOL Challenge Q28) Bread dough rises because the yeast mixed into it breaks down sugar without using oxygen and gives off a gas that makes bubbles in the dough. Which process is this, and which gas?

(A) Photosynthesis; oxygen
(B) Fermentation (anaerobic respiration); carbon dioxide
(C) Aerobic respiration; oxygen
(D) Fermentation; nitrogen

3. (LT 3.3 · SOL Challenge Q29) Two identical muscle cells each break down one molecule of glucose. One has plenty of oxygen; the other has none. How do their results compare?

(A) The cell with oxygen captures far more usable energy from the glucose
(B) Neither cell can release any energy from glucose
(C) Both cells capture the same energy, because the glucose is the same
(D) The cell without oxygen captures more energy, because it works faster

4. (LT 3.1 · Practice OB-U03-22) Think of ATP as a charged battery and ADP as a used one. In an animal cell, what supplies the energy to turn ADP back into ATP?

(A) Energy released by breaking down glucose in cellular respiration
(B) Energy from light absorbed in the chloroplast
(C) Energy released when the cell takes in water
(D) Energy released when carbon dioxide enters the cell

5. (LT 3.2 · Practice OB-U03-31) Which molecule completes the equation for photosynthesis?

$6CO_2 + 6H_2O + \text{light energy} \rightarrow _____ + 6O_2$

(A) ATP
(B) $C_6H_{12}O_6$ (glucose)
(C) CO_2 (carbon dioxide)
(D) Chlorophyll

6. (LT 3.3 · Practice OB-U03-33) A carrot's orange root grows underground and never receives light. Its cells are alive and use energy every day. How do the root cells get the ATP they need?

(A) They make sugar by photosynthesis in the dark
(B) They break down sugar sent down from the leaves by cellular respiration in their mitochondria
(C) They absorb ATP from the soil
(D) They do not need ATP because they are underground

7. (LT 3.4 · Practice OB-U03-41) Each fall, oak leaves pile up on the floor of a Prince William Forest Park hollow. By summer most of the pile is gone. What happened to most of the carbon in those leaves?

(A) It sank into the soil and turned into rock
(B) Fungi and bacteria broke the leaves down by cellular respiration and released the carbon as carbon dioxide
(C) It was destroyed, because carbon atoms are used up when leaves rot
(D) Sunlight turned it into oxygen

3. Draw the loop from memory

Draw a chloroplast and a mitochondrion. Use arrows to show what each one takes in and gives off, where the light comes in, and where the ATP goes. Label at least six arrows.

4. Fix the claim

A classmate writes: *"Plants do photosynthesis during the day and respiration only at night."* Rewrite the claim so it is correct, then give one piece of evidence from Unit 3 that supports your version.

5. Fix one

Pick a question in part 2 that you missed or guessed. What idea makes the right answer right?

6. My plan

Before my test or reassessment, the one thing I will review is:

Check questions 1–3. Scan, tap your answer and read why.
Bio Tool #8 · SOL Challenge Q27–Q29

Check questions 4–7. Scan, tap your answer and read why.
Bio Tool #8 · Unit 3 Practice

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